

Exp No: 1

Study of various Network commands used in Linux and Windows

AIM:

To study and practise the various network commands used in Linux and Windows operating systems.

ALGORITHM / PROCEDURE:

1. Open the Terminal in Linux and the Command Prompt (cmd) in Windows.
2. Use `hostname` to see the name of the computer.
3. Use `ifconfig` (Linux) and `ipconfig` (Windows) to see the IP address, subnet mask and MAC address.
4. Use `ping` to check whether the remote system is reachable. Note the packet loss and the time taken.
5. Use `tracert` (Linux) and `tracert` (Windows) to see the path taken by the packets.
6. Use `nslookup` and `dig` to find the IP address of a domain name.
7. Use `netstat` to see the active connections and open ports.
8. Use `arp -a` to see the IP address to MAC address table.
9. Use `route` (Linux) and `route print` (Windows) to see the routing table.
10. Note down the output of each command.

Commands used:

Purpose	Linux	Windows
Host name	<code>hostname</code>	<code>hostname</code>
IP configuration	<code>ifconfig / ip addr</code>	<code>ipconfig /all</code>
Check connectivity	<code>ping -c 4 google.com</code>	<code>ping google.com</code>
Trace the route	<code>tracert google.com</code>	<code>tracert google.com</code>
DNS lookup	<code>nslookup / dig</code>	<code>nslookup</code>
Active connections	<code>netstat -tuln</code>	<code>netstat -an</code>
ARP table	<code>arp -a</code>	<code>arp -a</code>
Routing table	<code>route -n</code>	<code>route print</code>
MAC address	<code>ip link</code>	<code>getmac</code>

DIAGRAM:

Fig 1: Network commands and the layers they work in

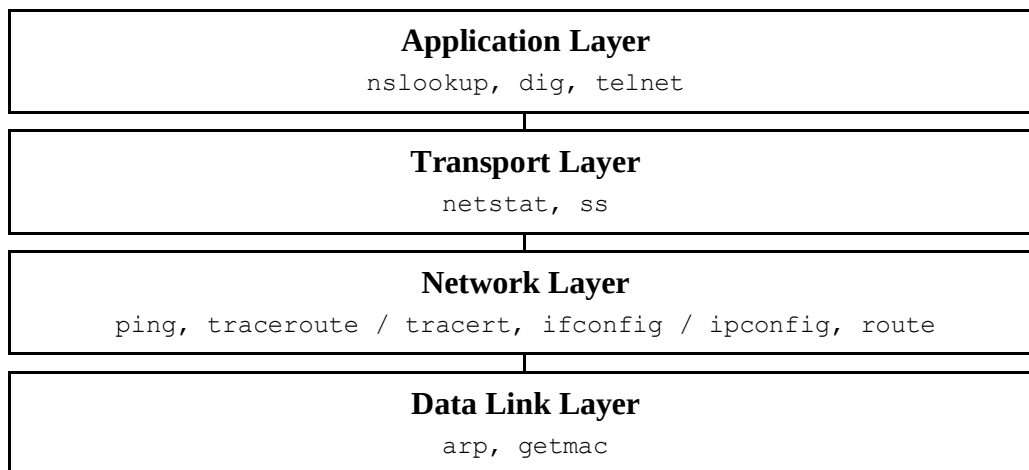
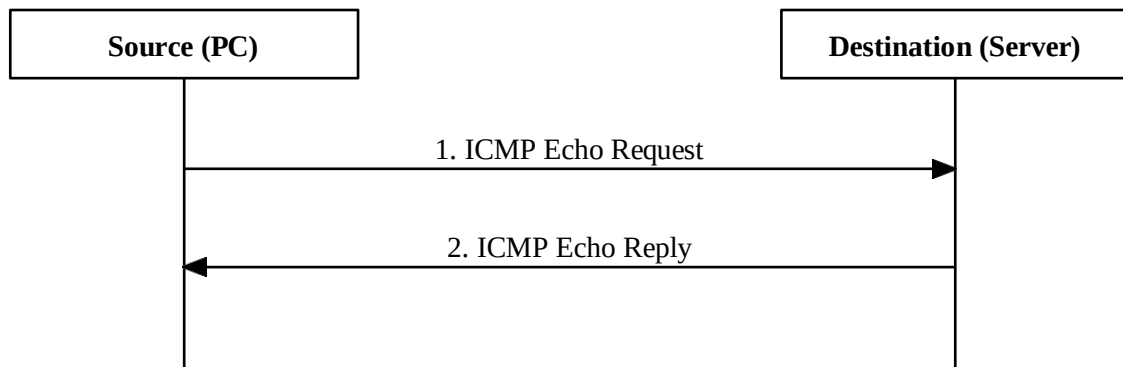


Fig 2: Working of ping command



OUTPUT:

Linux:

```
uSsEr @uSsErRr -VirtualBox ~
```

```
uSsEr @uSsErRr -VirtualBox:~$ hostname  
uSsEr -VirtualBox
```

```
uSsEr @uSsErRr -VirtualBox ~
```

```
uSsErRr @uSsErRr -VirtualBox:~$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255  
inet6 fe80::a00:27ff:fe4e:66a1 prefixlen 64 scopeid 0x20<link>  
ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)  
RX packets 2543 bytes 2154321 (2.1 MB)  
TX packets 1987 bytes 312456 (312.4 KB)  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
inet 127.0.0.1 netmask 255.0.0.0
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ ping -c 4 google.com
PING google.com (142.250.195.78) 56(84) bytes of data.
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=117 time=14.8 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=117 time=13.9 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=117 time=15.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 13.904/14.525/15.230/0.512 ms
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ traceroute google.com
traceroute to google.com (142.250.195.78), 30 hops max, 60 byte packets
 1 _gateway (192.168.1.1) 1.102 ms 1.045 ms 0.987 ms
 2 10.20.0.1 (10.20.0.1) 4.321 ms 4.287 ms 4.410 ms
 3 172.16.5.9 (172.16.5.9) 8.765 ms 8.801 ms 8.730 ms
 4 72.14.215.85 (72.14.215.85) 12.543 ms 12.498 ms 12.610 ms
 5 maa03s45-in-f14.1e100.net (142.250.195.78) 14.230 ms 14.187 ms 14.301 ms
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ nslookup google.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.78
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ dig google.com +short
142.250.195.78
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp6       0      0 :::80                  :::*                   LISTEN
udp        0      0 0.0.0.0:68             0.0.0.0:*
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ arp -a
_gateway (192.168.1.1) at c4:e9:84:2a:5b:10 [ether] on enp0s3
? (192.168.1.12) at 3c:52:82:7d:11:9e [ether] on enp0s3
```

UuSSErR @UuSSErR ~VirtualBox ~

```
UuSSErR @UuSSErR ~VirtualBox:~$ route -n
Kernel IP routing table
Destination Gateway      Genmask         Flags Metric Ref    Use Iface
0.0.0.0     192.168.1.1  0.0.0.0         UG    100    0      0 enp0s3
192.168.1.0 0.0.0.0     255.255.255.0   U     100    0      0 enp0s3
```

Windows:

Command Prompt

```
C:\Users\UuSsEeRr >hostname  
DESKTOP-UuSsEeRr
```

Command Prompt

```
C:\Users\UuSsEeRr>ipconfig /all  
  
Windows IP Configuration  
  
Host Name . . . . . : DESKTOP-UuSsEeRr  
Primary Dns Suffix . . . . . :  
Node Type . . . . . : Hybrid  
IP Routing Enabled. . . . . : No  
  
Ethernet adapter Ethernet:  
  
Physical Address. . . . . : 00-1A-2B-3C-4D-5E  
DHCP Enabled. . . . . : Yes  
IPv4 Address. . . . . : 192.168.1.25(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.1.1  
DHCP Server . . . . . : 192.168.1.1  
DNS Servers . . . . . : 8.8.8.8
```

Command Prompt

```
C:\Users\UuSsEeRr >ping google.com  
  
Pinging google.com [142.250.195.78] with 32 bytes of data:  
Reply from 142.250.195.78: bytes=32 time=15ms TTL=117  
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117  
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117  
Reply from 142.250.195.78: bytes=32 time=16ms TTL=117  
  
Ping statistics for 142.250.195.78:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

Command Prompt

```
C:\Users\UuSsEeRr >tracert google.com  
  
Tracing route to google.com [142.250.195.78]  
over a maximum of 30 hops:  
  
 0  <1 ms    <1 ms    <1 ms    192.168.1.1  
 1   4 ms     4 ms     4 ms     10.20.0.1  
 2   9 ms     8 ms     9 ms     172.16.5.9  
 3  13 ms    12 ms    12 ms    72.14.215.85  
 4  14 ms    14 ms    15 ms    142.250.195.78  
  
Trace complete.
```

Command Prompt

```
C:\Users\UuSsEeRr>nslookup google.com  
Server: dns.google  
Address: 8.8.8.8  
  
Non-authoritative answer:  
Name: google.com  
Address: 142.250.195.78
```

Command Prompt

```
C:\Users\UuUsSeErR>netstat -an
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	192.168.1.25:50234	142.250.195.78:443	ESTABLISHED
UDP	0.0.0.0:5353	*:*	

Command Prompt

```
C:\Users\UuUsSeErR>arp -a
```

```
Interface: 192.168.1.25 --- 0x8
```

Internet Address	Physical Address	Type
192.168.1.1	c4-e9-84-2a-5b-10	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static

Command Prompt

```
C:\Users\UuUsSeErR>route print
```

IPv4 Route Table

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.25	25
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
	192.168.1.0	255.255.255.0	On-link	192.168.1.25	281

Command Prompt

```
C:\Users\UuUsSeErR>getmac
```

Physical Address	Transport Name
------------------	----------------

00-1A-2B-3C-4D-5E	\Device\NPF{8F3A21B0-5C6D-4E7F-9A1B-2C3D4E5F6A7B}
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RESULT:

Thus the various network commands used in Linux and Windows were studied and executed successfully.